Hikmet Burak Özcan, Ph.D.

Mathematics and Statistics Tutor · IB · AP

İzmir, Türkiye · Online and in person
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Ph.D. Mathematics (2026) · 7 years teaching in English · IB AA/AI SL/HL · AP Calculus AB/BC · AP Statistics · TI-Nspire CX

Profile

Mathematics Ph.D. (İzmir Institute of Technology, 2026) with seven years of university teaching in calculus, probability and statistics, all in English. I tutor IB Diploma Mathematics, AP Calculus and AP Statistics one-to-one, from first concepts to final exam papers. Lessons focus first on why each method works, then on exam technique: mark schemes, timing and calculator use.

Mathematics and Statistics Expertise

Area	Key topics	Exam courses
Calculus	Limits, derivatives, integrals, series, differential equations	IB AA SL/HL, AP Calculus AB/BC
Probability and statistics	Distributions (binomial, Poisson, normal), sampling, confidence intervals, hypothesis and chi-square tests, regression	AP Statistics, IB AI SL/HL, IB AA
Algebra, functions and proof	Functions, sequences, trigonometry, vectors, complex numbers, induction	IB AA SL/HL, AP Precalculus

Also: IB Paper 3 (HL), IB Internal Assessment and Extended Essay guidance, SAT Math, A Level Mathematics.

Teaching Experience

Private Tutor – IB Mathematics, AP Calculus and AP Statistics Online and in person

- One-to-one lessons in IB Diploma Mathematics, AP Calculus and AP Statistics.
- Each student starts with a short diagnostic test, then follows a study plan built on past papers and mark schemes, with regular progress updates for students and parents.

Research and Teaching Assistant May 2019 – present
Department of Mathematics, İzmir Institute of Technology

- Lead weekly Probability and Statistics tutorials on random variables, distributions, estimation and hypothesis testing – the core of AP Statistics and IB statistics topics.
- Lead weekly tutorials for Calculus I–III, Basic Calculus, Calculus for Engineering and Science, Analytic Geometry and Fundamentals of Mathematics – the core of IB AA HL and AP Calculus BC.
- Write practice problems, run exam review sessions and grade exams, all in English; coordinated the department seminar series.

Talks: 14 seminar and conference talks (2018–2025).

Education

- **Ph.D. in Mathematics**, İzmir Institute of Technology, 2026. Thesis: *On Regularity of Hardy–Littlewood Maximal Operators*.
- **M.Sc. in Mathematics**, Dokuz Eylül University, 2020. Thesis: *Prime Ideal Theorem on Number Fields*.
- **B.Sc. in Mathematics**, İzmir University of Economics, 2016. GPA 3.57/4.00, ranked first in the class.

Tutoring Approach

Find the gaps with a short diagnostic test → **build understanding** of why each method works → **practise like the exam** with past papers, mark schemes, timed sections and fast GDC use → **track progress** with monthly goals and short updates for students and parents.

Skills and Research

Languages: Turkish (native), English (professional working; teaching in English since 2019), Russian (beginner).

Tools: TI-Nspire CX graphing calculator (GDC) for IB Papers 2 and 3 and AP exams; R and Python for statistics; LaTeX; Manim.

Lessons: online or in person in İzmir.

Selected publications: *The Green–Tao theorem and the infinitude of primes in domains*, The American Mathematical Monthly, 2022. *Rings with few units and the infinitude of primes*, Hacettepe J. Math. Stat., 2020.